

Joseph A. DePaolo-Boisvert

Chicago, IL • (860) 736-7977 • jdepaoloboisvert@hawk.iit.edu • joseph184depaolo@gmail.com

josephdepaoloboisvert.github.io • linkedin.com/in/jdepaolo-boisvert • github.com/josephdepaoloboisvert

SUMMARY

Computational chemist with six years of structure-based and machine-learning drug discovery across GPCR and bacterial membrane protein targets. Designed and executed a 4.2-million-compound virtual screen whose enumeration cycles produced pharmacophore hypotheses for an active medicinal chemistry collaboration; first author on the molecular dynamics study establishing structure-activity relationships for a novel antibiotic target; co-developer of an open-source enhanced-sampling suite now standard across his research group. Co-founder and CTO of a GPCR-focused drug discovery company. Runs projects independently at national supercomputing scale, and trains others to do the same.

EDUCATION

Ph.D., Chemistry — Illinois Institute of Technology, Chicago, IL

Conferral August 2026

Successfully defended April 2026. Advisor: Dr. David D. L. Minh. Master's-level requirements completed May 2022.

Dissertation: "Membrane Protein Molecular Dynamics: Application, Representation, Enhanced Sampling."

B.A., Chemistry and B.A., Physics — Boston University, Boston, MA

May 2020

TECHNICAL SKILLS

Computer-Aided Drug Design: Molecular docking and virtual screening at multi-million-compound scale (AutoDock Vina, MGLTools); iterative hit expansion and focused library enumeration; pharmacophore identification; protein-ligand interaction fingerprinting (ProLIF); binding pose clustering; alchemical and machine-learning binding free energy calculation (YANK); ligand polarization corrections; structure-guided SAR development.

Molecular Dynamics & Enhanced Sampling: OpenMM; parallel-tempering replica exchange with adaptive state insertion; MBAR equilibration detection and importance resampling (pyMBAR); membrane protein system construction (CHARMM-GUI, PDBFixer); structure repair and homology modeling (MODELLER, AlphaFold); protonation (PDB2PQR, AmberTools); force field parameterization and concatenation — ff14SB, Lipid17, OPC3/OPC, GAFF2, OpenFF/SMIRNOFF, AM1-BCC charge assignment via antechamber.

Machine Learning: JAX, Flax; stochastic and deterministic autoencoders; normalizing flows (RealNVP); nonlinear dimensionality reduction and intrinsic dimensionality estimation; HDBSCAN, UMAP, PCA, scikit-learn. ACS-NVIDIA Certificate of Machine Learning Knowledge.

Cheminformatics & Data: RDKit (conformer generation, maximal common substructure alignment, protonation), OpenFF Toolkit, SMILES handling; mining and curation from the Protein Data Bank, PubChem, GPCRdb, and Enamine libraries; MDTraj, MDAnalysis, NumPy, SciPy, Pandas, Matplotlib, Jupyter, NetCDF.

High-Performance Computing: Bridges2 (Pittsburgh Supercomputing Center), Expanse (San Diego Supercomputing Center), Jetstream2 (Indiana University); SLURM, PBS, and VM-based environments; CPU-parallel docking campaigns via batch partitioning and multiprocessing; GPU-accelerated MD; Python (expert), Bash, C; Git/GitHub.

EXPERIENCE

Co-Founder & Chief Technology Officer — Biagon Inc., Chicago, IL

Feb 2024 – Present

Physics-based and machine-learning workflows for GPCR-targeted drug discovery; spun out of the Minh laboratory at Illinois Tech.

- Co-founded the company with members of the Minh research group; write and test production code for the simulation and discovery pipeline and contribute to proprietary machine learning workflows.
- Advanced the company through Loyola I-Corps, the IIT Kaplan Institute Accelerator, and Nucleate; IP licensing negotiations with Illinois Tech ongoing.
- Engage CROs and prospective partners in business development, including at BIO annually since 2024; selected for the Babson College SUD Sprint (2025 cohort).

Doctoral Researcher — Illinois Institute of Technology, Department of Chemistry

Aug 2020 – Aug 2026

Advisor: Dr. David D. L. Minh. Projects run independently with biweekly reporting to faculty collaborators.

Virtual Screening & Structure-Based Antibacterial Discovery — NQR

- Designed and executed a **4.2-million-compound virtual screen** of the Enamine library against the NQR ubiquinone binding site on the Bridges2 supercomputer, partitioning the library into 50,000-compound batches distributed across full CPU nodes with one AutoDock Vina job per core.
- Advanced top-scoring hits through successive enumeration cycles — promoting leading compounds to scaffolds for focused libraries with systematic substitutions — surfacing pharmacophore hypotheses taken up by an inter-institutional medicinal chemistry collaboration (Becker, Loyola University; Juárez and Tuz, Illinois Tech).
- Contributed docking and molecular dynamics analysis to an FDA-approved drug repurposing effort against NQR, in collaboration with a medicinal chemistry group at Loyola University.
- Produced the first MD simulations of *Vibrio cholerae* NQR in complex with ubiquinone-4 and three potent inhibitors (HQNO, aurachin-D42, korormicin-A): 13 production simulations of ~350–400k atoms each, run in triplicate for ≥500 ns in a physiologically realistic heterogeneous membrane.
- Identified conserved hydrogen-bonding and hydrophobic binding motifs via interaction fingerprinting, distance analysis, and pose clustering, corroborated against published mutagenesis — establishing the structure-activity basis for rational antibiotic design. **First author, Proteins.**

ChiMPSS — Open-Source Membrane Protein Simulation Infrastructure

- Co-developed the Chicago Membrane Protein Simulation Suite, a three-repository open-source pipeline for reproducible enhanced-sampling MD of membrane proteins. **First author on MotorRow** (multi-stage equilibration); co-author on Bridgeport (automated preparation, repair, embedding, and ligand parameterization) and FultonMarket (adaptive PTRE execution).
- Implemented parallel-tempering replica exchange with dynamic temperature-state insertion driven by inter-state mixing probability, MBAR equilibration detection, and importance resampling; validated across four class-A GPCRs (CB2, 5HT2A, 5HT2B, KOR) with triplicate microsecond-scale ensembles converging from both common and structurally distinct starting configurations.
- Built a pan-GPCR reference frame from 555 GPCRdb structures; identified transducer-stabilization bias in crystallographic reference sets as a systematic confound for pharmacological inference.
- ChiMPSS is now the standard pipeline of the Minh group, in use across CB2, 5HT2A, 5HT2B, B2A, DOR, and μ opioid receptor splice variants. **Onboarded three graduate students** onto the codebase.

Machine Learning for Conformational Ensembles

- Developed stochastic autoencoders in JAX/Flax for nonlinear dimensionality reduction of MD trajectories; benchmarked systematically across five systems of increasing complexity (oxycodone, deca-alanine, crambin, BRD4 with JQ1, HIV-1 protease with peptide ligand) against PCA and deterministic autoencoding.
- Showed stochastic models resolve more distinct conformational clusters at low latent dimensionality and generalize better at the highest compression ratios; characterized intrinsic compression thresholds using nearest-neighbor dimensionality estimation, HDBSCAN, and UMAP.
- Contributed simulation and analysis to machine learning models relating GPCR intracellular pocket conformations to G protein and β -arrestin-2 signaling efficacy, computing efficacy along multiple pathways rather than binary activity classification (*J. Chem. Inf. Model.*). Separately quantified protein-induced ligand polarization as a corrective term in binding free energy calculations.

Computing Allocations & Infrastructure Leadership

- Graduate PI on ACCESS allocation CIS250014 and principal manager of an inherited XSEDE-era allocation; authored allocation applications on behalf of Dr. Oscar Juárez and renewal reports on behalf of Dr. David Minh across nine allocations supporting the thesis work. Applications subsequently adopted as templates by other students securing their own awards.
- Consumed tens of thousands of GPU-hours across national cyberinfrastructure; work supported by NIH R01AI151152, NIH R01GM127712, NSF Award #2331458, and Walder Foundation Grant 23-01215.

TEACHING & MENTORSHIP

Instructor of Record — CHEM 467, Medicinal Chemistry, Illinois Institute of Technology

Fall 2025

- Sole instructor for an upper-division medicinal chemistry lecture course. Designed the full curriculum — biological macromolecules as drug targets, pharmacokinetics and metabolism, and drug discovery, optimization, and regulation — plus a fourteen-paper primary literature syllabus, all assessment (two midterms, weekly problem sets, student journal clubs, and a capstone analysis of an FDA-approved drug), and an external guest lecture series.

Teaching Assistant — Illinois Institute of Technology

Aug 2020 – Aug 2026

- Six semesters of instruction, Undergraduate Physical Chemistry Laboratory; two semesters grading Undergraduate Biochemistry.
- Taught and helped train a physical chemistry laboratory student who moved into undergraduate research in the Minh group; he is now a shared co-author, ChiMPSS co-developer, and CEO of Biagon Inc.

Instructor — Molecular Simulation and Calculation Workshop, Barranquilla, Colombia

Mar 2022

- Taught binding free energy calculations and force field parameterization to an international cohort of researchers.

PUBLICATIONS & SOFTWARE

1. **DePaolo-Boisvert, J. A.**; Tuz, K.; Minh, D. D. L.; Juárez, O. X. "Molecular Dynamics Analysis of Inhibitor Binding Interactions in the *Vibrio cholerae* Respiratory Complex NQR." *Proteins* **2025**, 94 (2), 649–659.
2. **DePaolo-Boisvert, J. A.**; Willow, S. Y.; Srinarmwong, T.; Bahl, A.; Minh, D. D. L. "Deep Neural Network Auto-Encoding Models Reveal Inherent Dimensionality of Protein Structural Ensembles." *Manuscript in preparation*.
3. Cooper, D. A.; **DePaolo-Boisvert, J.**; Nicholson, S. A.; Gad, B.; Minh, D. D. L. "Intracellular Pocket Conformations Determine Signaling Efficacy through the μ Opioid Receptor." *J. Chem. Inf. Model.* **2025**, 65 (3), 1465–1475.
4. Sorescu, J. M.; González-Montalvo, M. A.; Yuan, M.; **DePaolo-Boisvert, J.**; et al.; Tuz, K.; Juárez, O. X. "Breakthroughs in the Development of Antibiotics, Antifungals and Antiparasitics Targeting the Pathogens' Respiratory Chain." *Crit. Rev. Biochem. Mol. Biol.* **2025**, 60 (4–6), 141–174.
5. González-Montalvo, M. A.; Sorescu, J. M.; Yuan, M.; **DePaolo-Boisvert, J.**; Liang, P.; Juárez, O. X.; Tuz, K. "NQR as a Target for New Antibiotics." *Front. Microbiol.* **2025**, 16, 1690572.
6. Tuz, K.; Yuan, M.; Hu, Y.; Do, T. T. T.; Willow, S. Y.; **DePaolo-Boisvert, J. A.**; Fuller, J. R.; Minh, D. D. L.; Juárez, O. "Identification of the Riboflavin Cofactor-Binding Site in the *Vibrio cholerae* Ion-Pumping NQR Complex." *J. Biol. Chem.* **2022**, 298 (8), 102182.

Software: MotorRow (first author), Bridgeport, and FultonMarket — github.com/CCBatIIT

SELECTED PRESENTATIONS

Oral — "Energy-Preserving Variational Auto-Encoding and Decoding of Atomistic Protein Structure." ACS National Meeting, San Francisco (Aug 2023); Gibbs 37 Conference on Biothermodynamics (Oct 2023).

Poster — "Solvation Free Energy Calculations using RealNVP Normalizing Flows." ACS National Meeting, Chicago (Aug 2022); Gibbs 36 (Oct 2022). Annual graduate research presentations, Illinois Tech (2021–2025).